# Finite Elements for Differential Geometry

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# Introduction

This lecture on *Finite Elements for Differential Geometry* aims to derive and discuss models, finite elements, and numerical methods for problems arising in (discrete) differential geometry, with a particular focus on applications in continuum mechanics.

The first part is devoted to problems involving extrinsic differential geometry, where the computational domain is embedded in the surrounding Euclidean space, e.g., a plate or surface in 3D. To solve nonlinear shell problems, we first discuss the equations of elasticity (continuum mechanics). It will turn out that this approach has significant weaknesses when one domain dimension is significantly smaller than the others. Thus, we perform dimension reduction, leading to (linear) beams and plates, and reveal several typical discretization problems for those problem classes. Non-standard finite elements like vector-valued tangential-continuous Nédélec, and normal-continuous Raviart–Thomas/Brezzi–Douglas–Marini elements as well as matrix-valued normal-normal-continuous Hellan–Herrmann–Johnson and tangential-tangential continuous Regge elements are motivated, constructed, and analyzed to construct robust discretization methods. Before going to shells, the "prima-donna" of structures, tools from so-called extrinsic differential geometry are repeated, where a surface is embedded in $\mathbb{R}^3$. We introduce the surface's fundamental forms and related quantities (e.g., metric and Weingarten tensor), (covariant) derivatives, and finite elements on surfaces. The discrete (distributional) curvature approximation is discussed. Combining knowledge of differential geometry and finite elements, we derive, discuss, and simulate nonlinear shells that deform under external forces.

In the second part of this lecture, we concentrate on the theory of Riemannian manifolds and the computation and approximation of their intrinsic quantities. Therefore, we do not assume that these manifolds are embedded. In terms of curvature, although the outer normal vector is not available anymore, the Gauss and geodesic curvature can still be computed. The approximation of the metric tensor using solely tangential-tangential continuous Regge finite elements, and the question of computing geodesics and the distributional Gauss curvature, will be fundamental. Again, we use tools of (discrete) differential geometry, such as the Gauss–Bonnet theorem, to define, analyze, and implement the resulting methods.

Hands-on materials and numerical examples can be found on GitHub:

https://github.com/MichaelNeunteufel/2026_FE_DiffGeo

These examples can be run directly in the browser without any installations required.

# Chapter 1

# Elasticity

This section introduces the equations of continuum mechanics (elasticity) to describe how an elastic body gets deformed in the presence of forces like gravity. For literature on (nonlinear) elasticity and continuum mechanics see e.g. [4, 7, 9, 2, 6].

## 1.1 Strain tensors

Let $\Omega \subset \mathbb{R}^d$ be an open and bounded domain in $d = 2, 3$ dimensions and let the boundary $\partial\Omega$ be sufficiently smooth. Then $\overline{\Omega}$ describes the reference configuration of a body, also called the undeformed configuration. The boundary is split into a Dirichlet and Neumann part $\Gamma_D$ and $\Gamma_N$, respectively, with $\Gamma_D \cap \Gamma_N = \emptyset$ and $\overline{\Gamma_D} \cup \overline{\Gamma_N} = \partial\Omega$. Further, we assume that the measure of $\Gamma_D$ is not zero, i.e., $|\Gamma_D| \neq 0$.

Applying external forces to the body leads to *deformation* represented by the function

$$\begin{aligned} \Phi : \overline{\Omega} &\to \mathbb{R}^d \\ x &\mapsto \Phi(x), \end{aligned} \tag{1.1.1}$$

which can be split additively into the identity function and the *displacement* $u$

$$\Phi = \mathrm{id} + u. \tag{1.1.2}$$

Next, we introduce the *deformation gradient*

$$\boldsymbol{F} := \boldsymbol{I} + \nabla u = \nabla\Phi, \tag{1.1.3}$$

where $\boldsymbol{I}$ denotes the identity matrix. A deformation is called *permissible* if the *deformation determinant* $J := \det(\boldsymbol{F})$ is greater than zero, $J > 0$, which entails that the material is non-interpenetrable, i.e., the orientation is preserved and volume elements with positive measure have also positive measure after the deformation.

To measure the quadratic change of lengths of the deformation, the (right) *Cauchy–Green strain tensor* is introduced as

$$\boldsymbol{C} := \boldsymbol{F}^\top \boldsymbol{F}. \tag{1.1.4}$$

It is also called *metric tensor* in the context of differential geometry and shells. There holds for $\Phi \in C^2(\Omega, \mathbb{R}^d)$ with Taylor's theorem for $\|\Delta x\| \to 0$

$$\begin{aligned} \frac{\|\Phi(x+\Delta x) - \Phi(x)\|^2}{\|\Delta x\|^2} &= \frac{\|\Phi(x) + \nabla\Phi\Delta x + \mathcal{O}(\|\Delta x\|^2) - \Phi(x)\|^2}{\|\Delta x\|^2} = \frac{\Delta x^\top \boldsymbol{F}^\top \boldsymbol{F} \Delta x + \mathcal{O}(\|\Delta x\|^3)}{\|\Delta x\|^2} \\ &= \frac{\Delta x^\top \boldsymbol{C} \Delta x}{\|\Delta x\|^2} + \mathcal{O}(\|\Delta x\|), \end{aligned} \tag{1.1.5}$$

i.e., it measures the change of lengths.

If $\boldsymbol{C} = \boldsymbol{I}$, the body does not get deformed, however, it could be rotated or translated. The following theorem states that these motions are precisely the kernel of $\boldsymbol{C} - \boldsymbol{I}$, so-called *rigid body motions* (or *Euclidean motion*).

**Theorem 1.1.1** *Let $\Omega$ be a connected domain and $\Phi \in C^1(\Omega, \mathbb{R}^d)$. Then a deformation is a rigid body motion if and only if*

$$\Phi \in RB := \{\Psi(x) = a + \boldsymbol{Q}x \,|\, a \in \mathbb{R}^d,\, \boldsymbol{Q} \in \mathrm{SO}(d)\}, \tag{1.1.6}$$

*where* $\mathrm{SO}(d) := \{\boldsymbol{A} \in \mathbb{R}^{d\times d} \,|\, \det \boldsymbol{A} = 1$ *and* $\boldsymbol{A}^{-1} = \boldsymbol{A}^\top\}$ *denotes the special orthogonal group.*

**Proof:** See e.g., [4, Theorem 1.8-1]. □

This motivates the definition of the *Green strain tensor*

$$\boldsymbol{E} := \frac{1}{2}(\boldsymbol{C} - \boldsymbol{I}) \tag{1.1.7}$$

measuring the real strains induced by the deformation $\Phi$. Its kernel is exactly the rigid body motions, $\ker \boldsymbol{E} = RB$. Inserting (1.1.3)–(1.1.4) yields the representation

$$\boldsymbol{E} = \frac{1}{2}\left(\nabla u^\top \nabla u + \nabla u^\top + \nabla u\right). \tag{1.1.8}$$

Assuming $\nabla u = \mathcal{O}(\varepsilon)$ with $1 \gg \varepsilon > 0$ small and neglecting all higher order terms gives the *linearized strain tensor*

$$\boldsymbol{\varepsilon}(u) := \frac{1}{2}\left(\nabla u^\top + \nabla u\right) = \operatorname{sym}(\nabla u) \tag{1.1.9}$$

used in linear elasticity.

**Remark 1.1.2 (Engineering notation)** *Objects in reference notation are frequently denoted by capital letters, e.g., $X$ for the position vector in the reference configuration. In contrast, objects in the deformed configuration are denoted by small letters, e.g., $x = \Phi(X)$ for the position vector in the deformed configuration. The deformation gradient is then written as $\boldsymbol{F} = \frac{\partial x}{\partial X}$. Volume elements are related by the deformation determinant $dv = J\,dV$. A line element $dx$ in deformed configuration can be expressed in reference configuration by $dx = \partial x/\partial X\,dX = \boldsymbol{F}\,dX$. So $dx^T dx = dX^T \boldsymbol{C} dX$ and we can define the Green strain tensor by relating the line elements $dx^T dx - dX^T dX = dX^T(\boldsymbol{C} - \boldsymbol{I})dX = 2dX^T\,\boldsymbol{E}dX$.*

## 1.2 Hyperelastic materials

We consider *hyperelastic constitutive relations*, i.e., the deformation energy is given by a potential $\Psi : \Omega \times \mathbb{R}^{d\times d} \to \mathbb{R}$

$$E_{\text{def}} = \int_\Omega \Psi(x, \boldsymbol{F}(u(x)))\,dx. \tag{1.2.1}$$

More general elastic materials and a derivation based on conservation laws can be found in the literature. Further, for ease of presentation, we assume that the potential is *homogeneous*, i.e.,

$$\Psi(x, \cdot) = \Psi(\cdot). \tag{1.2.2}$$

Given Dirichlet data $u_D$ on $\Gamma_D$, a static body load $f$, and traction forces $g$ on $\Gamma_N$ we can define the following minimization problem

$$\mathcal{W}(u) := \int_\Omega \Psi(\boldsymbol{F}(u)) - f \cdot u\,dx - \int_{\Gamma_N} g \cdot u\,ds \to \min!. \tag{1.2.3}$$

Gravity is a classical body force, whereas a car driving over a bridge is an example of a traction force.

We seek a function $u$ in the set of admissible displacements given by

$$U := \{u : \Omega \to \mathbb{R}^d \,|\, u = u_D \text{ on } \Gamma_D,\ \det(\boldsymbol{F}(u)) > 0\}. \tag{1.2.4}$$

Note that the constraint $\det(\boldsymbol{F}(u)) > 0$ is often given implicitly in the material law or is neglected in the small deformation case.

To compute the weak formulation of (1.2.3) we take the first variation in direction $\delta u$, well-known as the *principle of virtual works*, yielding

$$D_u\mathcal{W}(u)[\delta u] = \int_\Omega \frac{\partial \Psi}{\partial \boldsymbol{F}} : \nabla \delta u - f \cdot \delta u\,dx - \int_{\Gamma_N} g \cdot \delta u\,ds \overset{!}{=} 0 \tag{1.2.5}$$

for all admissible test functions $\delta u$, i.e., $\delta u \in U_{\Gamma_D}$, where

$$U_{\Gamma_D} := \{\delta u : \Omega \to \mathbb{R}^d \,|\, \delta u = 0 \text{ on } \Gamma_D\}. \tag{1.2.6}$$

In (1.2.5) we used the Frobenius scalar product of two matrices $\boldsymbol{A} : \boldsymbol{B} = \sum_{i,j=1}^d \boldsymbol{A}_{ij}\boldsymbol{B}_{ij}$ and the chain rule $D_u\Psi(\boldsymbol{F}(u))[\delta v] = \frac{\partial \Psi}{\partial \boldsymbol{F}}(\boldsymbol{F}(u)) : D_u(\boldsymbol{F}(u))[\delta u]$.

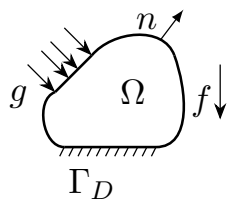


Figure 1.1: Initial configuration of a body $\Omega$ with Dirichlet boundary $\Gamma_D$, external forces $f$ and $g$, and outer unit normal vector $n$.

By defining the *first Piola–Kirchhoff stress tensor*

$$\boldsymbol{P} := \frac{\partial \Psi}{\partial \boldsymbol{F}} \tag{1.2.7}$$

and integration by parts in (1.2.5), we obtain the Euler–Lagrange equation of (1.2.3)

$$\begin{cases} -\mathrm{div}(\boldsymbol{P}) = f & \text{in } \Omega, \\ u = u_D & \text{on } \Gamma_D, \\ \boldsymbol{P}n = g & \text{on } \Gamma_N, \end{cases} \tag{1.2.8}$$

where $n$ denotes the outer normal vector of $\Omega$, cf. Figure 1.1.

**Definition 1.2.1** *We call a hyperelastic potential*

- objective *(*frame-indifferent*), if for all positive definite* $\boldsymbol{F} \in \mathbb{M}_+(d) := \{\boldsymbol{A} \in \mathbb{R}^{d\times d} \,|\, x^\top \boldsymbol{A} x > 0$ *for all* $0 \neq x \in \mathbb{R}^d\}$ *and* $\boldsymbol{Q} \in \mathrm{SO}(d)$

$$\Psi(\boldsymbol{Q}\boldsymbol{F}) = \Psi(\boldsymbol{F}). \tag{1.2.9}$$

- isotropic*, if for all* $\boldsymbol{F} \in \mathbb{M}_+(d)$ *and* $\boldsymbol{Q} \in \mathrm{SO}(d)$

$$\Psi(\boldsymbol{F}\boldsymbol{Q}) = \Psi(\boldsymbol{F}). \tag{1.2.10}$$

For objective materials, its response does not depend on the observer's frame of reference. In other words, the material depends only on stretches, not rotations (all real-life materials are objective). There are no preferred directions for an isotropic material, such as steel. Wood is a classical example of a strongly anisotropic material.

**Example 1.2.2** *The potential* $\Psi(\boldsymbol{F}) := \frac{\mu}{4}\left(\boldsymbol{F}^\top\boldsymbol{F} - \boldsymbol{I}\right) : \left(\boldsymbol{F}^\top\boldsymbol{F} - \boldsymbol{I}\right) + \frac{\lambda}{8}\,\mathrm{tr}(\boldsymbol{F}^\top\boldsymbol{F} - \boldsymbol{I})^2$*, with* $\mu > 0$*,* $\lambda \geq 0$ *is objective and isotropic (Exercise!).*

A crucial consequence of objectivity is that the energy potential only depends on the Cauchy–Green strain tensor.

**Theorem 1.2.3** *A potential* $\Psi : \mathbb{R}^{d\times d} \to \mathbb{R}$ *is frame-indifferent if and only if it is a function of the Cauchy–Green strain tensor* $\boldsymbol{C} = \boldsymbol{F}^\top\boldsymbol{F}$*, i.e.,*

$$\Psi(\boldsymbol{F}) = \hat{\Psi}(\boldsymbol{C}). \tag{1.2.11}$$

**Proof:** See e.g., [2, Chapter VI, Theorem 1.6] or [4, Theorem 3.3-1]. □

We will consider frame-indifferent potentials and write $\Psi$ independently of arguments $\boldsymbol{F}$, $\boldsymbol{C}$, and $\boldsymbol{E}$. Using Theorem 1.2.3, the weak form (1.2.5) can be rewritten as: Find $u \in U$ such that for all $v \in U_{\Gamma_D}$

$$\int_\Omega 2\frac{\partial \Psi}{\partial \boldsymbol{C}}(\boldsymbol{C}(u)) : \mathrm{sym}(\boldsymbol{F}^\top(u)\nabla v)\,dx = \int_\Omega f \cdot v\,dx + \int_{\Gamma_N} g \cdot v\,ds, \tag{1.2.12}$$

where we utilized that $D_u\boldsymbol{C}(u)[v] = D_u\big(\boldsymbol{F}(u)^\top\big)[v]\boldsymbol{F}(u) + \boldsymbol{F}(u)^\top D_u\big(\boldsymbol{F}(u)\big)[v] = 2\,\mathrm{sym}(\boldsymbol{F}^\top(u)\nabla v)$.

Defining the *second Piola–Kirchhoff stress tensor*

$$\boldsymbol{\Sigma} := 2\frac{\partial \Psi}{\partial \boldsymbol{C}} \tag{1.2.13}$$

and exploiting the symmetry of $\boldsymbol{\Sigma}$ gives

$$\int_\Omega \boldsymbol{F}\boldsymbol{\Sigma} : \nabla v\, dx = \int_\Omega f \cdot v\, dx + \int_{\Gamma_N} g \cdot v\, ds. \tag{1.2.14}$$

Consequently, we obtain the relation

$$\boldsymbol{P} = \boldsymbol{F}\boldsymbol{\Sigma}. \tag{1.2.15}$$

Note that in contrast to $\boldsymbol{\Sigma}$, $\boldsymbol{P}$ is not symmetric. For the sake of completeness, the symmetric *Cauchy stress tensor* $\boldsymbol{\sigma}$ acting on the deformed configuration of the body is introduced by

$$\boldsymbol{\sigma} = \frac{1}{J}\boldsymbol{F}\boldsymbol{\Sigma}\boldsymbol{F}^\top = \frac{1}{J}\boldsymbol{P}\boldsymbol{F}^\top = \frac{1}{J}\frac{\partial \Psi}{\partial \boldsymbol{F}}\boldsymbol{F}^\top. \tag{1.2.16}$$

Then by applying the transformation theorem and the chain rule ($v \circ \Phi^{-1} = \tilde{v}$, $(\nabla_x v) \circ \Phi^{-1} = \nabla_{\tilde{x}}\tilde{v}\boldsymbol{F}$)

$$\int_{\tilde{\Omega}} \frac{1}{J}\boldsymbol{F}\boldsymbol{\Sigma} : (\nabla_{\tilde{x}}\tilde{v}\boldsymbol{F})\, d\tilde{x} = \int_{\tilde{\Omega}} \boldsymbol{\sigma} : \nabla_{\tilde{x}}\tilde{v}\, d\tilde{x}.$$

Assuming a frame-indifferent and isotropic potential yields that the energy potential $\Psi$ is a function of the invariants of the Cauchy–Green strain tensor.

**Theorem 1.2.4 (Rivlin–Ericksen)** *Let $\mathbb{S}_>(d) := \{\boldsymbol{A} \in \mathbb{R}^{d\times d} \,|\, \boldsymbol{A}^\top = \boldsymbol{A}$ and $\det \boldsymbol{A} > 0\}$ denote the set of symmetric matrices with positive determinant. An energy potential $\Psi : \mathbb{R}^{d\times d} \to \mathbb{R}$ is frame-indifferent and isotropic if and only if there exists a function $\overline{\Psi} : \mathbb{S}_>(d) \to \mathbb{R}$ depending only on the invariants of the characteristic polynomial $\det(\lambda\boldsymbol{I} - \boldsymbol{C}) = \lambda^3 - I_1(\boldsymbol{C})\lambda^2 - I_2(\boldsymbol{C})\lambda - I_3(\boldsymbol{C})$ such that*

$$\Psi(\boldsymbol{F}) = \overline{\Psi}(I_1(\boldsymbol{C}), I_2(\boldsymbol{C}), I_3(\boldsymbol{C})) = \gamma_0(I_1, I_2, I_3)\boldsymbol{I} + \gamma_1(I_1, I_2, I_3)\boldsymbol{C} + \gamma_2(I_1, I_2, I_3)\boldsymbol{C}^2, \tag{1.2.17}$$

*where, with $\mathrm{tr}(\boldsymbol{C})$ and $\det(\boldsymbol{C})$ denoting the trace and determinant of $\boldsymbol{C}$,*

$$I_1(\boldsymbol{C}) = \mathrm{tr}(\boldsymbol{C}), \qquad I_2(\boldsymbol{C}) = \frac{1}{2}\left(\mathrm{tr}(\boldsymbol{C})^2 - \mathrm{tr}(\boldsymbol{C}^2)\right), \qquad I_3(\boldsymbol{C}) = \det(\boldsymbol{C}). \tag{1.2.18}$$

**Proof:** See e.g., [8] or [4, Theorem 3.6-1]. □

**Example 1.2.5** *From the Rivlin–Ericksen theorem the potential in the Green strain tensor is of the form $\Psi(\boldsymbol{E}) = \Psi(\mathrm{tr}(\boldsymbol{E}), \mathrm{tr}(\boldsymbol{E}^2), \det \boldsymbol{E})$. Assume that $\Psi$ has at $\boldsymbol{E} = 0$ a local minimum with value 0 and is smooth. Defining $\mu := \Psi_{,2}(\mathrm{tr}(\boldsymbol{E}), \mathrm{tr}(\boldsymbol{E}^2), \det \boldsymbol{E})|_{\boldsymbol{E}=0}$ and $\lambda := \Psi_{,11}(\mathrm{tr}(\boldsymbol{E}), \mathrm{tr}(\boldsymbol{E}^2), \det \boldsymbol{E})|_{\boldsymbol{E}=0}$, where $\Psi_{,2}(\cdot,\cdot,\cdot)$ denotes the partial derivative with respect to the second argument and $\Psi_{,11}(\cdot,\cdot,\cdot)$ two derivatives in the first one, there holds*

$$\Psi(\boldsymbol{E}) = \frac{\lambda}{2}\,\mathrm{tr}(\boldsymbol{E})^2 + \mu\,\boldsymbol{E} : \boldsymbol{E} + \mathcal{O}(\|\boldsymbol{E}\|^3).$$

Ignoring the higher-order terms we obtain the *St. Venant–Kirchhoff* material law

$$\Psi_{\mathrm{VK}}(\boldsymbol{E}) = \mu\|\boldsymbol{E}\|_F^2 + \frac{\lambda}{2}\,\mathrm{tr}(\boldsymbol{E})^2, \qquad \|\boldsymbol{A}\|_F^2 := \boldsymbol{A} : \boldsymbol{A}, \tag{1.2.19}$$

is widely used to model nonlinear behavior in a moderate deformation regime. However, it may happen for large deformations that elements get compressed heavily and are even pressed through others. The material law of *Neo–Hooke*

$$\Psi_{\mathrm{NH}}(\boldsymbol{C}) = \frac{\mu}{2}\left(\mathrm{tr}(\boldsymbol{C} - \boldsymbol{I}) - \log(\det(\boldsymbol{C}))\right) + \frac{\lambda}{8}(\log(\det(\boldsymbol{C})))^2 \tag{1.2.20}$$

Figure 1.2: Two solutions of rubber strip clamped on left and right boundary.

prevents the non-physical behavior of the St. Venant–Kirchhoff material law, as there holds

$$\Psi_{\mathrm{NH}}(\boldsymbol{C}) \to \infty \quad \text{for } \det(\boldsymbol{C}) = (\det(\boldsymbol{F}))^2 \to 0 \text{ or } \det(\boldsymbol{C}) \to \infty, \tag{1.2.21}$$

i.e., infinite energy is needed to completely compress or stretch the material. We remark that, especially in the (nearly) incompressible regime, i.e. $\lambda \gg 1$ and thus $\det \boldsymbol{C} \approx 1$, a slightly different material law of Neo–Hooke is also used, namely

$$\tilde{\Psi}_{\mathrm{NH}}(\boldsymbol{C}) = \frac{\mu}{2}\left(\operatorname{tr}(\boldsymbol{C}-\boldsymbol{I}) - \log(\det(\boldsymbol{C}))\right) + \frac{\lambda}{2}(\underbrace{\sqrt{\det(\boldsymbol{C})}}_{=\det \boldsymbol{F}} - 1)^2. \tag{1.2.22}$$

The two material constants $\mu > 0$ and $\lambda > 0$ used above are the so-called Lamé parameters. Two more physically interpretable constants are the Young's modulus $E > 0$, representing the stiffness of a solid material, and the Poisson's ratio $0 \leq \nu < 1/2$[1]. The latter describes the expansion or contraction in the perpendicular direction to the force compressing or stretching the material. Physical units are $[\mu] = [\lambda] = [E] = \frac{N}{m^2} = Pa$ Pascal, whereas the Poisson ratio is dimensionless. To convert these parameters, the following formulae are used

$$\nu = \frac{\lambda}{2(\lambda+\mu)}, \qquad E = \frac{\mu(3\lambda+2\mu)}{\lambda+\mu}, \tag{1.2.23a}$$

$$\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}, \qquad \mu = \frac{E}{2(1+\nu)}. \tag{1.2.23b}$$

Note that in the limit $\nu \to \frac{1}{2}$, or equivalently $\lambda \to \infty$, the material is called incompressible, necessitating special (numerical) treatment.

## 1.3 Theoretical results of nonlinear elasticity

The question of existence and uniqueness of this (highly nonlinear) equations is delicate. From physical examples, it is known that we cannot expect uniqueness. Considering e.g., a rubber strip with both ends fixed and neglecting gravity, the identity is a trivial solution for the displacement. When twisting one of the ends by $2\pi$ ($360°$), we obtain a different solution to the same boundary conditions, see Figure 1.2. In this manner, we can even produce infinitely many solutions by twisting. Other examples are given, e.g., by buckling (bifurcations) or snap-through problems, where two solutions exist for one external force (one before and one after the snap-through).

For existence, one approach exploits, if possible, the polyconvex structure of the energy potential $\Psi$ in its invariants $I_i(\boldsymbol{C})$, compare (1.2.18), proving the existence of minimizers [1]. The material law of St. Venant–Kirchhoff is not polyconvex, but the Neo–Hooke material law falls in this category. A large class of Ogden-type materials fulfills this assumption. Another ansatz uses the implicit function theorem, obtaining a unique solution for small data. Therefore, relatively strong regularity assumptions must be made [4].

Only a little rigorous mathematical numerical analysis for finite elasticity has been accomplished so far. E.g., in [3] a priori error estimates for finite element discretizations in nonlinear elasticity are discussed for polyconvex materials under the assumption of sufficiently small right-hand sides.

[1] For meta-materials, the Poisson ratio can also take negative values.

## 1.4 Linearized elasticity

Under the assumption of small deformations all three stress tensors (1.2.7), (1.2.13), and (1.2.16) reduce to one, which we denote by $\boldsymbol{\sigma}$ (e.g., $\boldsymbol{P} = \boldsymbol{F}\boldsymbol{\Sigma} = (\boldsymbol{I} + \nabla u)\boldsymbol{\Sigma} = \boldsymbol{\Sigma} + \mathcal{O}(\varepsilon^2)$ as $\nabla u = \mathcal{O}(\varepsilon)$ by assumption). Assuming a quadratic potential $\Psi(\cdot)$, we deduce a linear stress-strain relation by $\boldsymbol{\sigma} = \frac{\partial \Psi}{\partial \boldsymbol{\varepsilon}} = \mathcal{C}\boldsymbol{\varepsilon}$, where $\mathcal{C} \in \mathbb{R}^{d\times d\times d\times d}$ denotes the fourth-order elasticity tensor. For an three-dimensional isotropic and frame-indifferent material, the stress-strain relation can be written in Voigt notation as

$$\begin{pmatrix} \boldsymbol{\sigma}_{11} \\ \boldsymbol{\sigma}_{22} \\ \boldsymbol{\sigma}_{33} \\ \boldsymbol{\sigma}_{23} \\ \boldsymbol{\sigma}_{13} \\ \boldsymbol{\sigma}_{12} \end{pmatrix} = \frac{E}{(1+\nu)(1-2\nu)} \begin{pmatrix} 1-\nu & \nu & \nu & & & 0 \\ \nu & 1-\nu & \nu & & & \\ \nu & \nu & 1-\nu & & & \\ & & & 1-2\nu & & \\ & & & & 1-2\nu & \\ & 0 & & & & 1-2\nu \end{pmatrix} \begin{pmatrix} \boldsymbol{\varepsilon}_{11} \\ \boldsymbol{\varepsilon}_{22} \\ \boldsymbol{\varepsilon}_{33} \\ \boldsymbol{\varepsilon}_{23} \\ \boldsymbol{\varepsilon}_{13} \\ \boldsymbol{\varepsilon}_{12} \end{pmatrix}, \tag{1.4.1}$$

or in matrix notation $\boldsymbol{\sigma} = \frac{E}{(1+\nu)(1-2\nu)}((1-2\nu)\boldsymbol{\varepsilon} + \nu\,\mathrm{tr}(\boldsymbol{\varepsilon})\boldsymbol{I}) = 2\mu\boldsymbol{\varepsilon} + \lambda\,\mathrm{tr}(\boldsymbol{\varepsilon})\boldsymbol{I}$. For $\nu \neq \frac{1}{2}$ relation (1.4.1) can be inverted, $\boldsymbol{\varepsilon} = \mathcal{C}^{-1}\boldsymbol{\sigma}$,

$$\begin{pmatrix} \boldsymbol{\varepsilon}_{11} \\ \boldsymbol{\varepsilon}_{22} \\ \boldsymbol{\varepsilon}_{33} \\ \boldsymbol{\varepsilon}_{23} \\ \boldsymbol{\varepsilon}_{13} \\ \boldsymbol{\varepsilon}_{12} \end{pmatrix} = \frac{1}{E} \begin{pmatrix} 1 & -\nu & -\nu & & & \\ -\nu & 1 & -\nu & & & \\ -\nu & -\nu & 1 & & & \\ & & & 1+\nu & & \\ & & & & 1+\nu & \\ & & & & & 1+\nu \end{pmatrix} \begin{pmatrix} \boldsymbol{\sigma}_{11} \\ \boldsymbol{\sigma}_{22} \\ \boldsymbol{\sigma}_{33} \\ \boldsymbol{\sigma}_{23} \\ \boldsymbol{\sigma}_{13} \\ \boldsymbol{\sigma}_{12} \end{pmatrix}, \tag{1.4.2}$$

and $\mathcal{C}^{-1}$ is called the *compliance tensor*. Relation (1.4.2) can be written compactly as

$$\boldsymbol{\varepsilon} = \frac{1}{E}\left((1+\nu)\boldsymbol{\sigma} - \nu\ \mathrm{tr}(\boldsymbol{\sigma})\boldsymbol{I}\right) = \frac{1}{E}\left((1+\nu)\,\mathrm{dev}(\boldsymbol{\sigma}) + \frac{1-2\nu}{3}\,\mathrm{tr}(\boldsymbol{\sigma})\boldsymbol{I}\right), \tag{1.4.3}$$

where $\mathrm{dev}(\boldsymbol{A}) = \boldsymbol{A} - \frac{1}{d}\,\mathrm{tr}(\boldsymbol{A})\boldsymbol{I}$ denotes the deviatoric (trace-free) part of a matrix $A \in \mathbb{R}^{d\times d}$. Here, the case $\nu = \frac{1}{2}$ is well-defined leading to the identity $\boldsymbol{\varepsilon} = \frac{1+\nu}{E}\,\mathrm{dev}(\boldsymbol{\sigma})$, which is unique up to the trace of $\boldsymbol{\sigma}$.

The strong form of (1.2.8) becomes in the linearized case

$$\begin{cases} -\mathrm{div}(\boldsymbol{\sigma}) = f & \text{in } \Omega, \\ u = u_D & \text{on } \Gamma_D, \\ \boldsymbol{\sigma} n = g & \text{on } \Gamma_N, \end{cases} \tag{1.4.4}$$

and the variational problem (inserting the stress-strain relation) reads: Find $u \in U$ such that for all $v \in U_{\Gamma_D}$

$$\int_\Omega \mathcal{C}\boldsymbol{\varepsilon}(u) : \boldsymbol{\varepsilon}(v)\,dx = \int_\Omega f \cdot v\,dx + \int_{\Gamma_N} g \cdot v\,ds. \tag{1.4.5}$$

The material laws of St. Venant–Kirchhoff and Neo–Hooke reduce to the linear material law of Hooke

$$\Psi_{\mathrm{H}}(\boldsymbol{\varepsilon}) := \mu\|\boldsymbol{\varepsilon}\|_F^2 + \frac{\lambda}{2}(\,\mathrm{tr}(\boldsymbol{\varepsilon}))^2 \tag{1.4.6}$$

and we obtain the problem: Find $u \in U$ such that for all test functions $v \in U_{\Gamma_D}$

$$\int_\Omega 2\mu\,\boldsymbol{\varepsilon}(u) : \boldsymbol{\varepsilon}(v) + \lambda\,\mathrm{div}(u)\mathrm{div}(v)\,dx = \int_\Omega f \cdot v\,dx + \int_{\Gamma_N} g \cdot v\,ds. \tag{1.4.7}$$

The unique solvability of (1.4.5) is proven in the next section. Note that (1.4.5) and (1.4.7) coincide for $\mathcal{C}$ defined as in (1.4.1).

## 1.5 Analysis of linear elasticity

We use the classical Lax–Milgram Lemma for elliptic problems to prove existence and uniqueness for the linear elasticity problem (1.4.7).

**Lemma 1.5.1 (Lax–Milgram)** *Let $a(\cdot,\cdot) : V \times V \to \mathbb{R}$ be a continuous and elliptic bilinear form with constants $\alpha$ and $\beta$, respectively. Then, for all $f \in V^*$ there exists a unique solution $u \in V$ of the problem*

$$a(u,v) = f(v), \qquad \text{for all } v \in V \tag{1.5.1}$$

*and there holds the stability estimate*

$$\|u\|_V \le \frac{1}{\beta}\|f\|_{V^*}. \tag{1.5.2}$$

Surprisingly, the entire gradient of a vector-valued function can be bounded by its symmetric part, cf. [5], at the cost of possibly large constants for anisotropic domains.

**Lemma 1.5.2 (Korn's inequality)** *Let $\Omega$ be a connected and bounded Lipschitz domain (i.e., the boundary can locally be parameterized by a Lipschitz function). Then there exists a constant $\hat{c}_K > 0$ such that*

$$\hat{c}_K^2 \|u\|_{H^1}^2 \le \|u\|_{L^2}^2 + \|\boldsymbol{\varepsilon}(u)\|_{L^2}^2 \tag{1.5.3}$$

*for all $u \in H^1(\Omega, \mathbb{R}^d)$ and $\hat{c}_K$ depends only on $\Omega$. Assume further that $\Gamma_D \subset \partial\Omega$ has positive measure. Then there exists a constant $c_K > 0$ such that*

$$c_K^2 \|u\|_{H^1}^2 \le \|\boldsymbol{\varepsilon}(u)\|_{L^2}^2 \tag{1.5.4}$$

*for all $u \in H^1_{\Gamma_D}(\Omega, \mathbb{R}^d)$. The constants $\hat{c}_K$ and $c_K$ tend to zero for deteriorating aspect ratio.*

Korn's inequality relies on the fact that by setting Dirichlet boundary conditions at parts of the boundary, the kernel of the symmetric gradient operator consisting of linearized rigid body motions

$$\ker \boldsymbol{\varepsilon}(\cdot) = RB^{\text{lin}} := \{\Phi(x) = \boldsymbol{A}\,x + b \,|\, b \in \mathbb{R}^d,\ \boldsymbol{A} \in \mathbb{R}^{d\times d} \text{ with } \boldsymbol{A}^\top = -\boldsymbol{A}\} \tag{1.5.5}$$

is locked.

**Corollary 1.5.3** *The bilinear form*

$$a(u,v) := \int_\Omega 2\mu\, \boldsymbol{\varepsilon}(u) : \boldsymbol{\varepsilon}(v) + \lambda\, \mathrm{div} u\, \mathrm{div} v\, dx \tag{1.5.6}$$

*is continuous and coercive (elliptic) with*

$$|a(u,v)| \le (2\mu + \lambda)\, \|u\|_{H^1}\, \|v\|_{H^1}\,,$$
$$a(u,u) \ge 2\mu\, c_K^2\, \|u\|_{H^1}^2\,.$$

Céa's lemma yields the following a-priori finite element error estimate for Lagrange elements $u_h \in U_h \subset U$

$$\|u - u_h\|_{H^1}^2 \le \frac{2\mu + \lambda}{2\mu\, c_K^2} \inf_{v_h \in U_h} \|u - v_h\|_{H^1}^2\,, \tag{1.5.7}$$

from which we deduce that the problem is ill conditioned if the material is nearly incompressible, $\lambda \gg \mu$, or the geometry is anisotropic leading a small Korn constant, $c_K \ll 1$.

**Exercise 1.5.4**

1. *Proof Corollary 1.5.3 and that the right-hand sides* (1.4.7) *are continuous functionals in $H^1(\Omega, \mathbb{R}^d)$.*
2. *Show that Korn's inequality is sharp by constructing a specific displacement on $\Omega = (0,1) \times (-t/2, t/2)$ and $U = \{u \in H^1(\Omega, \mathbb{R}^2) \,|\, u(0,\cdot) = 0\}$ and showing that $c_K \le c\,t$. Hint: Find $u$ such that $\|\boldsymbol{\varepsilon}(u)\|^2/\|\nabla u\| \le ct^2$.*
3. *Define the pressure-like variable $p = \lambda\, \mathrm{div} u$ as an additional field and rewrite the equations into a Stokes system (with penalty term). Argue that by using a Stokes stable pairing for $(u,p)$, nearly incompressible elasticity can be simulated robustly.*

# Chapter 2

# Timoschenko and Bernoulli beam

We saw in the previous chapter that for strong anisotropic structures the elasticity problem is ill-posed (due to Korn's inequality) and numerical simulations showed so-called locking behavior. Discretizing thin-walled structures with isotropic (small) elements is inefficient due to the vast number of elements needed. Using elements reflecting the structure and aspect ratio would be more economical, but lead to (shear) locking behavior. For structures, where one dimension is of magnitudes smaller than the other directions, it is expected that the material's behaviour in this direction can be described with fewer parameters, such that a dimension reduction to the mid-surface is an attractive strategy. One has to mesh and discretize objects of one dimension less, leading to smaller matrices. In addition to the computational speed-up, it was hoped that the locking behavior of full elasticity would be cured. However, as we will see, the shear locking problem is shifted but not circumvented or replaced with the difficulties of solving a fourth-order (biharmonic) problem. Nevertheless, the resulting beam equations yield more insight into the issue, which will be helpful when going to plates and shells.

## 2.1 Dimension reduction to beams

If we have a structure $\Omega \subset \mathbb{R}^2$ where one dimension is significantly thinner than the other, a dimension reduction to a one-dimensional problem can be done. If the two-dimensional structure is of the form $\Omega = (0,1) \times (-t/2, t/2)$ we can perform a reduction to the mid-surface $\mathcal{S} = (0,1) \times \{0\}$. We assume for simplicity that Dirichlet boundary conditions are prescribed on the left and right boundary $\{0,1\} \times (-t/2, t/2)$ and homogeneous Neumann boundaries ("do-nothing condition") at the top and bottom.

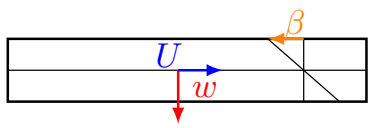


Figure 2.1: Horizontal deflection $U$, vertical deflection $w$, and rotation of normal $\beta$ of a beam.

To derive the beam equation in the setting of linear elasticity, we can start with a Galerkin-semidiscretization by approximating an infinite sum in the thickness direction

$$u = (u^x, u^y) \approx \left( \sum_{i=0}^{N_x} u_i^x(x) y^i, \sum_{i=0}^{N_y} u_i^y(x) y^i \right). \tag{2.1.1}$$

We consider the lowest possible order, where the (linearized) rigid body motions, i.e. both translations and a rotation, are included, leading to $N_x = 1$, $N_y = 0$ and

$$u = (U(x) - y\,\beta(x), w(x)). \tag{2.1.2}$$

Here, $U(x)$ and $w(x)$ correspond to the horizontal and vertical deflection, respectively, and $\beta(x)$ is the linearized rotation of the normal vector on the initial configuration, cf. Figure 2.1. Analogously, test functions are given by $v = (V(x) - y\delta(x), v(x))$. Inserting into the equation of linear elasticity (1.4.7) we obtain

$$\varepsilon(u) = \begin{pmatrix} U' - y\beta' & \frac{1}{2}(w' - \beta) \\ \frac{1}{2}(w' - \beta) & 0 \end{pmatrix}, \qquad \varepsilon(v) = \begin{pmatrix} V' - y\delta' & \frac{1}{2}(v' - \delta) \\ \frac{1}{2}(v' - \delta) & 0 \end{pmatrix},$$

$$\varepsilon(u) : \varepsilon(v) = (U' - y\beta')(V' - y\delta') + \frac{1}{2}(w' - \beta)(v' - \delta), \qquad \operatorname{div}(u)\operatorname{div}(v) = (U' - y\beta')(V' - y\delta')$$

and further

$$\begin{aligned} a(u,v) &= \int_0^1 \int_{-t/2}^{t/2} 2\mu\, \varepsilon(u) : \varepsilon(v) + \lambda \operatorname{div}(u)\operatorname{div}(v)\, dy\, dx \\ &= \int_0^1 \int_{-t/2}^{t/2} (2\mu + \lambda)(U' - y\beta')(V' - y\delta') + \frac{\mu}{2}(w' - \beta)(v' - \delta)\, dy\, dx \\ &= \int_0^1 (2\mu + \lambda) t\, U'V' + (2\mu + \lambda)\frac{t^3}{12}\beta'\delta' + \frac{t\,\mu}{2}(w' - \beta)(v' - \delta)\, dx. \end{aligned}$$

Assuming that the force is independent of the thickness direction $f = (f^x(x), f^y(x))$ the elasticity problem decouples into a Poisson-like *membrane problem*: Find $U \in H_0^1((0,1))$ such that for all $V \in H_0^1((0,1))$

$$(2\mu + \lambda)t \int_0^1 U'V' \, dx = t \int_0^1 f^x V \, dx \tag{2.1.3}$$

and the *bending problem*: Find $(w, \beta) \in H_0^1((0,1)) \times H_0^1((0,1))$ such that for all $(v, \delta) \in H_0^1((0,1)) \times H_0^1((0,1))$

$$(2\mu + \lambda)\frac{t^3}{12} \int_0^1 \beta'\delta' \, dx + \frac{t\,\mu}{2} \int_0^1 (w' - \beta)(v' - \delta) \, dx = t \int_0^1 f^y v \, dx. \tag{2.1.4}$$

Neglecting the Lamé parameters and constants and rescaling the right-hand side $f := \frac{1}{t^2} f^y$ we obtain the *Timoshenko beam* (also called Timoshenko–Ehrenfest beam): Find $(w, \beta) \in H_0^1((0,1)) \times H_0^1((0,1))$ such that for all $(v, \delta) \in H_0^1((0,1)) \times H_0^1((0,1))$

$$\int_0^1 \beta'\delta' \, dx + \frac{1}{t^2} \int_0^1 (w' - \beta)(v' - \delta) \, dx = \int_0^1 f\, v \, dx, \tag{2.1.5}$$

which corresponds to the minimization problem

$$\mathcal{W}(w, \beta) := \frac{1}{2} \int_0^1 (\beta')^2 \, dx + \frac{1}{2t^2} \int_0^1 (w' - \beta)^2 \, dx - \int_0^1 f\, w \, dx \to \min!. \tag{2.1.6}$$

The first term is the beam's bending energy, which measures how strongly the beam curves. The second term is the shearing energy, which measures how much the initial configuration's deformed normal deviates from the deformed configuration's normal (given by $w'$).

The Timoshenko beam involves a parameter, the thickness $t$, so the bending and shearing parts are scaled differently. Although the Timoshenko beam is $H_0^1 \times H_0^1$ elliptic, its constant depends on the thickness, which leads with Céa's Lemma to a parameter dependent a-priori error estimate

$$\|(w_h, \beta_h) - (w, \beta)\|_X \le c(t) \inf_{(v_h, \delta_h) \in X_h} \|(v_h, \delta_h) - (w, \beta)\|_X. \tag{2.1.7}$$

In the limit of vanishing thickness, $t \to 0$, it acts as a penalty enforcing the constraint $\beta = w'$. Therefore, in the limit, we can eliminate the rotation $\beta$ leading to the fourth-order minimization problem

$$\mathcal{W}(w) = \frac{1}{2} \int_0^1 (w'')^2 \, dx - \int_0^1 f\, w \, dx \to \min!, \tag{2.1.8}$$

which corresponds to the variational formulation: Find $w \in H_0^2((0,1))$ such that for all $v \in H_0^2((0,1))$

$$\int_0^1 w''v'' \, dx = \int_0^1 f\, v \, dx. \tag{2.1.9}$$

This fourth-order ODE problem is called the *Euler–Bernoulli beam*.

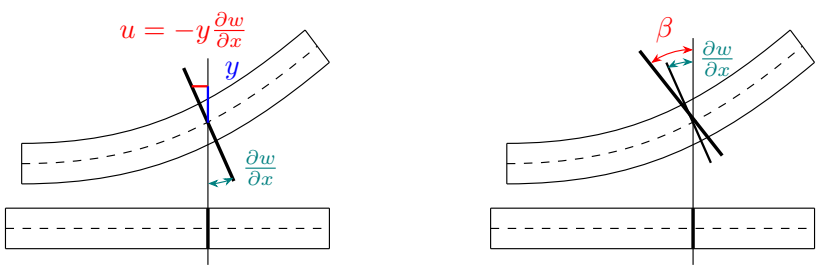


Figure 2.2: Rotation of Euler–Bernoulli beam and Timoshenko beam.

## 2.2 Discretization of Timoshenko beam

A straightforward discretization is to use Lagrange finite elements of the same order for the displacement and rotation $(w_h, \beta_h) \in U_h^k \times U_h^k$, which we will see may induce locking for a small thickness parameter $t$. In the limit $t \to 0$, the shearing energy can be seen as a penalty enforcing $w_h' = \beta_h$. In the lowest-order case, this identity forces the piece-wise constant $w_h'$ to fit with the linear and continuous $\beta_h$, leading to the trivial solution, $w_h = \beta_h = 0$. From the theory of mixed methods, we will see that inserting an $L^2$-projection into the shearing term cures this problem by relaxing the constraint. In 1D, this is equivalent to a numerical under-integration using only the mid-point rule for the shearing term. Using for $w_h$ one polynomial order higher than $\beta_h$ improves the locking behavior. This, however, leads to sub-optimal (linear) convergence rates for $\beta$. In Section 2.4 we analyze and present a shear-locking-free (discretized) method for the Timoshenko beam.

We used clamped boundary conditions in the derivation, $u, \beta \in H_0^1((0,1))$. Other boundary conditions like free, simply supported, or constrained rotation are possible, define $Q := \frac{1}{t^2}(w' - \beta)$ and $M := \beta'$:

| boundary condition | $w$ | $\beta$ | |
|---|---|---|---|
| clamped | D | D | $w = 0,\ \beta = 0$ |
| simply supported | D | N | $w = 0,\ M = 0$ |
| free | N | N | $Q = 0,\ M = 0$ |
| constraint rotation | N | D | $Q = 0,\ \beta = 0$ |

**Exercise 2.2.1** *Derive, implement, and test the different boundary conditions for the Timoshenko beam. Motivate the physical quantities $Q$ and $M$.*

In the limit $t \to 0$, the equality $w' = \beta$ holds. Therefore, the rotation $\beta$ has to coincide with the linearized normal vector of the deformed beam and can be eliminated from the equation, leading to the Euler–Bernoulli beam.

## 2.3 Discretization of Euler–Bernoulli beam

In contrast to the Timoshenko beam, the thickness parameter $t$ does not enter the Euler–Bernoulli beam equation anymore. Therefore, it does not suffer from locking, but $C^1$-conforming finite elements are required, as we now face a fourth-order problem. In 1D, constructing $C^1$-elements is easy by e.g. Hermite polynomials. This task is much more involved for higher dimensions, especially if unstructured (triangular) grids are required. As pre-work for plates and shells, we focus on formulations enabling Lagrangian elements for the vertical deflection $w$.

Let us rewrite the Euler–Bernoulli beam as a mixed saddle-point problem by introducing the linearized moment $\sigma = w''$ as an additional unknown

$$\int_0^1 \sigma\tau\,dx \qquad - \int_0^1 w''\tau\,dx = 0 \qquad \text{for all } \tau, \tag{2.3.1a}$$

$$-\int_0^1 \sigma\, v''\,dx \qquad = -\int_0^1 f\,v\,dx \qquad \text{for all } v, \tag{2.3.1b}$$

and use integration by parts such that all terms are well defined for $(w, \sigma) \in H_0^1((0,1)) \times H^1((0,1))$

$$\int_0^1 \sigma\tau\,dx \ + \int_0^1 w'\tau'\,dx = 0 \qquad \text{for all } \tau \in H^1((0,1)), \tag{2.3.2a}$$

$$\int_0^1 \sigma' v'\,dx \qquad = -\int_0^1 f\,v\,dx \qquad \text{for all } v \in H_0^1((0,1)). \tag{2.3.2b}$$

We can use Lagrange finite elements for discretization. The following section shows that this leads to a stable formulation and method.

**Exercise 2.3.1** *Show that $\sigma \in H^1((0,1))$ together with $w \in H_0^1((0,1))$ leads to clamped boundary conditions, $w = w' = 0$. What changes if $\sigma \in H_0^1((0,1))$? How can we incorporate free boundary conditions at $x = 1$ and clamped boundary conditions at $x = 0$?*

## 2.4 Analysis of Timoshenko and Euler–Bernoulli beam

For analyzing mixed saddle-point problems, the Lax–Milgram Lemma cannot be applied due to the missing coercivity on the whole product space. Brezzi's Theorem [2, 4.3 Theorem, 4.11 Theorem] gives sufficient conditions to prove well-posedness for this kind of problem.

### 2.4.1 Brezzi's Theorem for saddle point problems

Indefinite saddle-point problems, arising e.g., for minimization problems under constraints, are of the following general form: Find $(u,p) \in V \times Q$ such that for all $(v,q) \in V \times Q$

$$a(u,v) + b(v,p) = f(v), \tag{2.4.1a}$$
$$b(u,q) \qquad\qquad = g(q). \tag{2.4.1b}$$

Equation (2.4.1b) enforces a constraint on $u$, which can be incorporated as a penalty formulation yielding the structure ($t > 0$ small)

$$a(u,v) + b(v,p) \;= f(v), \tag{2.4.2a}$$
$$b(u,q) - t\,c(p,q) = g(q). \tag{2.4.2b}$$

**Theorem 2.4.1 (Brezzi)** *Assume that $a(\cdot,\cdot) : V \times V \to \mathbb{R}$ and $b(\cdot,\cdot) : V \times Q \to \mathbb{R}$ are continuous bilinear forms, i.e.,*

$$|a(u,v)| \le \alpha_2 \|u\|_V \|v\|_V \quad \textit{for all } u,v \in V, \tag{2.4.3}$$
$$|b(u,q)| \le \beta_2 \|u\|_V \|q\|_Q \quad \textit{for all } u \in V, \textit{ for all } q \in Q. \tag{2.4.4}$$

*Assume there holds coercivity of $a(\cdot,\cdot)$ on the kernel, i.e.,*

$$a(u,u) \ge \alpha_1 \|u\|_V^2 \quad \textit{for all } u \in V_0, \tag{2.4.5}$$
$$V_0 := \{u \in V \mid b(u,q) = 0 \quad \textit{for all } q \in Q\} \tag{2.4.6}$$

*and there holds the Ladyzhenskaya–Babuška–Brezzi (LBB) condition*

$$\sup_{u \in V} \frac{b(u,q)}{\|u\|_V} \ge \beta_1 \|q\|_Q \quad \textit{for all } q \in Q. \tag{2.4.7}$$

*Then, the mixed problem* (2.4.1) *is uniquely solvable. The solution fulfills the stability estimate*

$$\|u\|_V + \|p\|_Q \le c\,(\|f\|_{V^*} + \|g\|_{Q^*}) \tag{2.4.8}$$

*with the constant $c$ depending on $\alpha_1, \alpha_2, \beta_1$, and $\beta_2$.*

**Theorem 2.4.2 (extended Brezzi)** *Assume all requirements of Theorem 2.4.1 are fulfilled. Further, let $c(\cdot,\cdot)$ be a continuous and non-negative bilinear form and $a(\cdot,\cdot)$ be non-negative. Then, for $t \le 1$, the mixed problem* (2.4.2) *has a unique solution, fulfilling the following stability estimate independent of $t$*

$$\|u\|_V + \|p\|_Q \le c\,(\|f\|_{V^*} + \|g\|_{Q^*})\,, \qquad c \ne c(t). \tag{2.4.9}$$

### 2.4.2 Analysis of Timoshenko beam

We analyze the Timoshenko beam as mixed formulation by defining the shear $\gamma = \frac{1}{t^2}(w' - \beta)$ as additional unknown: Find $(w,\beta,\gamma) \in H_0^1((0,1)) \times H_0^1((0,1)) \times L^2((0,1))$ such that for all $(v,\delta,\xi) \in H_0^1((0,1)) \times H_0^1((0,1)) \times L^2((0,1))$

$$\int_0^1 \beta'\delta'\,dx \qquad + \int_0^1 (v' - \delta)\gamma\,dx = \int_0^1 f\,v\,dx, \tag{2.4.10a}$$
$$\int_0^1 (w' - \beta)\xi\,dx - t^2 \int_0^1 \gamma\,\xi\,dx \qquad = 0. \tag{2.4.10b}$$

Note that we multiplied (2.4.10b) with $t^2$ such that the thickness now appears in the numerator. The limit case $t = 0$ is therefore well-defined.

**Proposition 2.4.3** *The mixed formulation of the Timoshenko beam* (2.4.10) *is uniquely solvable and stable.*

**Proof:** Continuity of the bilinear forms and non-negativity of $a(\cdot,\cdot)$ and $c(\cdot,\cdot)$ are apparent.

On the kernel

$$\begin{aligned} B_0 &= \Big\{(w,\beta) \in H_0^1((0,1)) \times H_0^1((0,1)) \,\Big|\, \int_0^1 (w'-\beta)\gamma\,dx = 0 \text{ for all } \gamma \in L^2((0,1))\Big\} \\ &= \{(w,\beta) \in H_0^1((0,1)) \times H_0^1((0,1)) \,|\, w' = \beta\} \end{aligned}$$

there holds with Friedrichs's inequality[1]

$$\|\beta'\|_{L^2}^2 = \frac{1}{2}|\beta|_{H^1}^2 + \frac{1}{2}|\beta|_{H^1}^2 \geq c\left(\|\beta\|_{H^1}^2 + \|\beta\|_{L^2}^2\right) \geq c\left(\|\beta\|_{H^1}^2 + \|w\|_{H^1}^2\right). \tag{2.4.11}$$

Further, following [2], for given $\gamma \in L^2((0,1))$ define $\rho(x) := x(1-x)$ and

$$A := \int_0^1 \gamma(s)\,ds \Big/ \int_0^1 \rho(s)\,ds, \quad w(x) := \int_0^x \gamma(s)\,ds - A\int_0^x \rho(s)\,ds, \quad \beta(x) := -A\rho(x).$$

Then $\rho, w, \beta \in H_0^1((0,1))$ and there holds with a constant $c > 0$, $\|w'\|_{L^2} \leq c\|\gamma\|_{L^2}$, $\|\beta'\|_{L^2} \leq c\|\gamma\|_{L^2}$, and $w' - \beta = \gamma$. Thus, the LBB condition is fulfilled

$$\sup_{(w,\beta)\in H^1\times H^1} \frac{\int_0^1 (w'-\beta)\gamma\,dx}{\|w\|_{H^1} + \|\beta\|_{H^1}} \geq \frac{\|\gamma\|_{L^2}^2}{\|w'\|_{L^2} + \|\beta'\|_{L^2}} \geq \frac{1}{2c}\|\gamma\|_{L^2}. \tag{2.4.12}$$

Brezzi's Theorem 2.4.2 now finishes the proof, giving the robust estimate

$$\|\beta\|_{H^1} + \|w\|_{H^1} + \frac{1}{t^2}\|w' - \beta\|_{L^2} \leq c\|f\|_{H^{-1}}, \quad c \neq c(t). \tag{2.4.13}$$

□

Using Lagrange finite elements for $w$ and $\beta$ and one polynomial order less $L^2$-conforming discontinuous (i.e. piece-wise polynomials) elements for $\gamma$ gives a stable discretization for the Timoshenko beam, which can easily be shown (Exercise!). We obtain the optimal $t$-independent convergence rates:

**Corollary 2.4.4** *Let* $(w,\beta,\gamma)$ *be the solution of the continuous problem* (2.4.10) *and* $(w_h,\beta_h,\gamma_h) \in U_{h,0}^k \times U_{h,0}^k \times Q_h^{k-1}$ *(the space of continuous piece-wise polynomials of order* $k$ *and discontinuous piece-wise polynomials of order* $k-1$*) the corresponding discrete one for a positive integer* $k$*. If the exact solution is sufficiently smooth, there holds*

$$\|w - w_h\|_{H^1} + \|\beta - \beta_h\|_{H^1} + \|\gamma - \gamma_h\|_{L^2} \leq ch^k(\|w\|_{H^{k+1}} + \|\beta\|_{H^{k+1}} + \|\gamma\|_{H^k}), \qquad c \neq c(t). \tag{2.4.14}$$

The second equality (2.4.10b) states in the discrete case that the shearing can be expressed as $\gamma_h = \frac{1}{t^2}\Pi^{L^2,k-1}(w_h' - \beta_h)$. Inserting this into the first equation (2.4.10a) yields the problem: Find $(w_h,\beta_h) \in U_{h,0}^k \times U_{h,0}^k$ such that for all $(v_h,\delta_h) \in U_{h,0}^k \times U_{h,0}^k$

$$\int_0^1 \beta_h'\delta_h'\,dx + \frac{1}{t^2}\int_0^1 \Pi^{L^2,k-1}(w_h' - \beta_h)\Pi^{L^2,k-1}(v_h' - \delta_h)\,dx = \int_0^1 f\,v_h\,dx, \tag{2.4.15}$$

which corresponds to a numerical under-integration of the shear term. For example, in the lowest-order case, the projection into piece-wise constants can be realized using the mid-point rule for numerical integration.

This (non-standard) primal method is equivalent to the mixed formulation, so we obtain the same stability and convergence results. Its advantage is obtaining a positive definite stiffness matrix, whereas the mixed system leads to an indefinite matrix.

[1] Friedrich: $\|u\|_{H^1} \leq c_F|u|_{H^1}, \quad$ for all $u \in H_0^1(\Omega)$

**Remark 2.4.5** *When using at least quadratic polynomials for the vertical deflection and rotation no shear locking occurs for the Timoshenko beam anymore, however, convergence rates with a sub-optimal rate are obtained for $\beta$ in a pre-asymptotic regime depending on $t$. As we will see later for the Reissner–Mindlin plate (the 2D analog to the Timoshenko beam), its analysis and numerical treatment to avoid shear locking is more involved.*

**Remark 2.4.6** *Setting $t = 0$ in* (2.4.10b) *leads to a stable discretization for the Euler–Bernoulli beam.*

**Exercise 2.4.7** *Show with Remark 2.4.6, that the solution of the Timoshenko beam $(w, \beta, \gamma)^{\mathrm{TB}}$ converges to the Euler–Bernoulli beam solution $(w, \beta, \gamma)^{\mathrm{EBB}}$*

$$\|(w, \beta, \gamma)^{\mathrm{TB}} - (w, \beta, \gamma)^{\mathrm{EBB}}\| \leq c\, t^2, \qquad c \neq c(t). \tag{2.4.16}$$

*Hint: Take the difference of the solutions as a new unknown for the difference of the corresponding equations and use Brezzi's Theorem.*

### 2.4.3 Analysis of Bernoulli beam

From the Lax–Milgram Lemma, we directly deduce that the Euler–Bernoulli beam is well-posed on $H_0^2((0,1))$ and Céa's Lemma yields a standard a-priori error estimate. The big drawback is that we need $C^1$ finite elements for a conforming discretization. In 1D, Hermite polynomials are directly applicable; however, extensions to 2D Kirchhoff–Love plates are complicated. Therefore, we analyze the mixed method (2.3.2) with Brezzi enabling the usage of Lagrange finite elements.

**Proposition 2.4.8** *The mixed formulation* (2.3.2) *of the Euler–Bernoulli beam yields a unique solution.*

**Proof:** By choosing $\sigma = w$, the LBB condition is fulfilled. The kernel

$$V_0 = \left\{ \sigma \in H^1((0,1)) \,\middle|\, \int_0^1 \sigma' w' = 0, \quad \text{for all } w \in H_0^1((0,1)) \right\} = \{\sigma = ax + b \,|\, a, b \in \mathbb{R}\}$$

consists only of linear functions. As all norms are equivalent in finite-dimensional spaces, the kernel coercivity is also fulfilled. All bilinear forms are continuous for the $H^1$-norm such that the conditions for Brezzi's theorem 2.4.1 are fulfilled. □

# Bibliography


[1] John M Ball. Convexity conditions and existence theorems in nonlinear elasticity. *Archive for Rational Mechanics and Analysis*, 63(4):337–403, 1976.

[2] Dietrich Braess. *Finite Elements: Theory, Fast Solvers, and Applications in Solid Mechanics.* Cambridge University Press, Cambridge, 3rd edition, 2007.

[3] Carsten Carstensen and Georg Dolzmann. An a priori error estimate for finite element discretizations in nonlinear elasticity for polyconvex materials under small loads. *Numerische Mathematik*, 97(1):67–80, 2004.

[4] Philippe G Ciarlet. *Mathematical Elasticity. Vol. I: Three-Dimensional Elasticity*, volume 20. North-Holland, Amsterdam, 1988.

[5] Georges Duvant and Jacques Louis Lions. *Inequalities in Mechanics and Physics*, volume 219 of *Grundlehren der mathematischen Wissenschaften.* Springer-Verlag, Berlin Heidelberg, 1 edition, 1976.

[6] Patrick Le Tallec. Numerical methods for nonlinear three-dimensional elasticity. volume 3 of *Handbook of Numerical Analysis*, pages 465–622. Elsevier, 1994.

[7] Jerrold E Marsden and T. J. R. Hughes. *Mathematical foundations of elasticity.* Dover Publication, Inc, New York, 1994.

[8] R. S. Rivlin and J. L. Ericksen. Stress-deformation relations for isotropic materials. *Journal of Rational Mechanics and Analysis*, 4:323–425, 1955.

[9] Peter Wriggers. *Nonlinear finite element methods.* Springer Science & Business Media, Heidelberg, 1 edition, 2008.